Chapter I

Introduction – The Emerging Interaction Society

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Introduction

Recently it has been argued that there is a need for computer science, and related fields of research, to shift its focus from user tasks, their requirements, applications or computing, to issues concerning interaction, mutual awareness, and ubiquity (e.g., Dourish, 2001). Overall, it is a shift from the *Information Society*, with its focus on information, storage and processing of data and transactions, to the *Interaction Society*, with related issues including, e.g., work as *ongoing and fluid networks of connections* (Sproull & Keisler, 1998) interaction overload (Ljungberg & Sørensen, 2000), interaction management (Whittaker et al., 1997), contact management (Whittaker et al., 2002), session management (Edwards, 1994), time management, etc. This general shift has also highlighted the need to acknowledge issues such as attention management in relation to the fluidity of work (e.g., Hudson et al., 2002; Davenport & Beck, 2001).

As acknowledged by Boden (1994), interaction is really the glue that builds up the modern organisation:

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> *"The structuring properties of the interaction order in real-time settings such as meetings have enormous (and as yet largely ignored) consequences for the overall structuring of organizations. Caught in a meeting and connected through a series of interactions across time and space are the people, ideas, decisions, and outcomes that make the organization" (Boden, 1994, p.106).*

With *interaction* and its related issues in mind, several empirical studies have been conducted with a specific focus on the characteristics of informal, lightweight, and opportunistic interaction and its implications for design of technology to support interaction (e.g., Whittaker et al., 1994; Wiberg, 2001a; Dahlberg et al., 2000). Together with the issue of *interaction* there is also the issue of *interruptions* and inappropriate times for interaction. Recent empirical work place studies (e.g., Hudson et al., 2002) have, for example, shown that people spend a lot of their time in settings where their ability to respond to incoming interaction requests are very limited.

The overall objective of this book is to provide its audience with a rich overview of this emerging *Interaction Society* enabled by new information and communication technologies (ICTs), such as mobile phones, PDAs, and pagers, and applications such as email and chat clients, instant messaging systems, video conferencing systems, and different kinds of alert- and notification systems.

A New Computer Supported Society is Emerging

A digital layer on top of our traditional society is emerging, i.e., a digital layer to the extent that modern information technology is used almost everywhere in today's society to support all various kinds of new technology-enabled human activities not possible to perform without the technology. Some researchers in the field have characterized the relation between these human activities and the enabling technologies in terms of IT dependencies where new human actions couldn't have been realized without the new technology (e.g., Nordstrom, 2003). One simple example of technology that has fast become widespread and adopted broadly over the whole human population is the mobile phone. With this small device a person can interact with somebody else independent of the geographic distance between them, and independent of their respective

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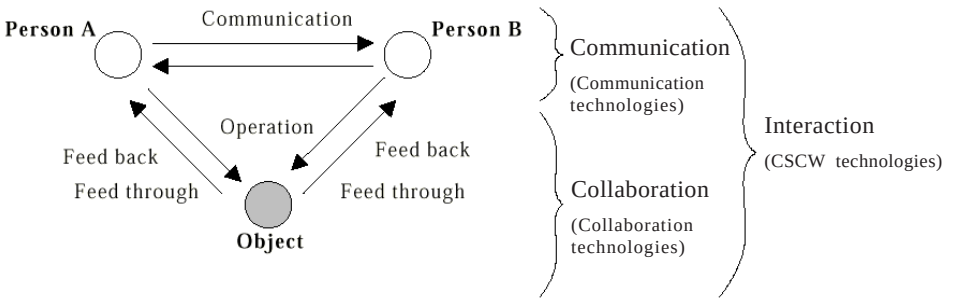
locations. Clearly this kind of new interaction device enables us to interact in new ways, and we would have some difficulties living without them today.

Before moving forward any farther, the very concept of "*interaction*" might need some clarification. Below I will therefore try to define this term in relation to the concepts of communication and collaboration.

The concepts of interaction and interaction support can be defined in relation to the concepts of communication and collaboration (Ljungberg, 1999). According to Dix & Beale (1996), communication is the exchange of information between people, e.g., video conferencing. Collaboration is when two or more people are operating a common object (or artifact), e.g., co-operative authoring where the shared document is the common object. In collaboration, operations produce "feedback" to the operator, but also "feed through" to co-workers. Support for collaboration is sometimes combined with support for communication, e.g., a collaborative authoring system (collaboration) equipped with a chat feature (communication). In the context of this model, communication and collaboration can be conceived as subsets of "interaction." As suggested by Ljungberg (1999) we can use "CSCW technologies" to frame the technological support for interaction, i.e., communication technologies and collaboration technologies (see Figure 1).

According to Figure 1, mobile interaction, which is a major component of the Interaction Society, can thus be defined as communication and collaboration around a shared object in a mobile use context, e.g., talking with somebody over a mobile phone while walking around on the deck on a big ship.

Figure 1. Definition of interaction and interaction support and how it relates to the concepts of communication and collaboration (Ljungberg, 1999; based on Dix & Beale, 1996)



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New IT Use: From Calculation to Communication

While the computer of yesterday was occupied with crunching numbers, today and tomorrow's technology will be occupied with maintaining our social contacts with one another. The computers of yesterday were commonly labeled as "information technologies," "IT" or "IS" (i.e., Information Systems). Now it might be a good idea to reinterpret this "IT" acronym. Due to the more social focus of modern IT use it might be more appropriate to talk about IT in terms of "Interaction Technologies" as a general label for what this technology does for us on a social level.

To be social is something very fundamental to us as humans and therefore it is quite easy to understand why these new technologies and devices to support human communication have so rapidly become adopted on a widespread basis. It is sometimes claimed that a loved baby has many names. That is also true for this new technology that has already been commonly labeled as, e.g., ICT (information and communication technology), social belonging technologies, awareness technologies, groupware, and virtual community support. These technologies appear nowadays everywhere and are used by a broad population ranging from adults to teenagers to children.

One important aspect of this new kind of IT use is the difference in the purpose of the technology, i.e., to support various kinds of social activities instead of supporting, e.g., advanced calculations. Another difference is in its use patterns. While computing in the 1970s was about several persons working together around *one* single machine to make *it* produce an *exact* result, today's computing is about *several* persons interacting with *each other* via *several* computers and, as such, *it* enables them to maintain and develop their social networks. Here, it becomes clear that also the *role* of the technology has changed from being in the frontline of our attention to now becoming a more pervasive technology that enable us to do new things without directing our attention to the technology *per se*. How many people think about how the phone operates "under the shell" while they are having a phone conversation?

There are several factors that together enable the *Interaction Society* including not only the new technical devices such as pagers, mobile phones and BlackBerries (i.e., mobile email devices, see Figure 2), but also, e.g., the underlying infrastructures (e.g., the Internet and the mobile phone network), new interaction modalities, new applications and services, and maybe fore-

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Figure 2. Two different BlackBerry mobile email devices



most, new use patterns and new emerging behaviors due to the adoption of these technologies.

In the next section we go through these different enabling factors and take a closer look at a few of the new behaviors that have started to arise around these interaction technologies.

Enabling Components for the Interaction Society

As stated in the previous section, the *Interaction Society* is enabled by several factors including, e.g., mobile devices, new applications, technical infrastructures, etc. These are all important factors that enable this new society to become a reality. Then, on top of these technologies, new use patterns and behaviors are emerging around these technologies. In this section the enabling technologies are first presented followed by a brief introduction of some new human behaviors that have already started to grow out of this new kind of IT use.

Interaction Infrastructures for the Interaction Society

From a technical viewpoint, the Interaction Society is mainly enabled on top of two major and global technical platforms, i.e., 1) *the telephone network*,

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(including mobile networks for voice and data communication), and 2) *the Internet.* These two platforms enable people who are geographically dispersed to communicate across the globe independent of the distance in between them. The telephone network enabled people to synchronously talk to one another. On the other hand, Internet enabled for more sustained interaction and enabled people to more or less asynchronously communicate via email, chat rooms, virtual communities, discussion forums, bulletin boards, etc. Today, these two platforms are now melting together into one global communication network where people can communicate with each other across these two previously separated infrastructures. A good example of the melting of these two platforms is the now growing market for IP telephony where the Internet is used to enable Internet users to make voice calls to other persons on the telephone network.

So, clearly these two platforms enable computerized interaction support. But, in order to reach the user and enable him or her to start to interact there are several additional components needed. Below some of these components are presented.

Interaction Devices

On top of the telephone network and the Internet infrastructure several new interaction devices have been developed to enable us to communicate at anytime, and from almost any location. Today, these technologies no longer tie us to the desktop or the office environment. Rather, the modern interaction devices are *mobile*, thus enabling us to take advantage of them in any preferable place and whenever we want to. From that perspective the technology has finally become ubiquitous and an integrated part of our lives (and not just an integrated part of our office environment). From a perceptual perspective, one of the most visible sign of today's Interaction Society might be the widespread adoption of mobile phones[1], and the second most visible sign of how computers are entering our everyday lives might be the widespread adoption of laptop computers and the everyday increasing numbers of Internet accounts[2].

However, it is not only laptop computers and mobile phones that count as interaction devices, and it has not only been efforts made on bridging *long distances* between geographically dispersed persons (like enabling communication over the Internet between different countries, or via the telegraph or the modern mobile phone), but there has also been efforts made on developing

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mobile devices to bridge *quite short distances* (i.e., the walkie-talkie supports one-to-one voice communication over two to three kilometers).

There have also been some efforts made on supporting human interaction across *really short distances*. One such example is the Japanese LoveGety (see Figure 3). The LoveGety is a matchmaking device that communicates with co-located devices in the near proximity of its user, i.e., around 10 meters. Owners can set the device to show display lights according to whatever mood they are in (there are only three): "let's just chat," "let's go sing some karaoke," or the "Get together" mode. The idea is to bring the device to, e.g., a night club or other social situation where others are equipped with similar devices and then do a virtual hide and seek until you find somebody that matches your distributed profile. When one LoveGety detects another of the opposite sex within range, it beeps and flashes green if both are in the same harmonious mode, and red if the opposite user is sending out a different mode.

In fact, there is a lot of research going on today that focuses on interaction support for co-located communities similar to the basic idea behind the commercial LoveGety device (e.g., supporting social gatherings and groups at fairs, conferences, rock concerts, etc.). During recent years several research prototype devices have been developed to support, e.g., mobile group awareness [i.e., the Hummingbird device (Holmqvist et al., 1999)], *ongoing mobile interaction* [i.e., the RoamWare system (Wiberg, 2001b), see Figure 4], *mobile meetings* [i.e., the ProxyLady system (Dahlberg et al., 2000)],

Figure 3. A picture of two LoveGetties (to the left a female device and to the right a male device)



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Figure 4. Two pictures of the RoamWare system (The first picture (left) shows the RoamWare device with a connection cable to allow for synchronization of data between the RoamWare device and a PDA or a PC. The second picture (right) shows a use situation where two persons are wearing the devices on their belts in a mobile meeting.)





mobile co-located communities [i.e., the Memetags system (Borovoy et al., 1998)], or even *mobile folklore* [i.e., "iBalls" (Borovoy et al., 2000)].

Figure 4 shows two pictures of the RoamWare system. The RoamWare system allows its users to keep a dynamic contact list that is automatically generated depending on who they meet during the day. This contact history can then be used to, e.g., send group emails, send meeting invitations or initiate a teleconference session with a group of persons that the user has just met, e.g., during a business lunch, a seminar or any other social activity where people gather together at the same physical place. As seen in the pictures in Figure 4, the RoamWare device does not have a graphic display, nor does it have any tiny keyboard or any other input modality. Rather, this is a "background technology" or a *calm technology* (Weiser & Brown, 1996) that is designed to only operate in the background of its user and let the user focus his or her attention the social context rather then on the mobile device *per se*.

Recently there have also been some commercial interaction devices developed with the same focus on supporting co-located groups and social gatherings. One such example is the Spot-me device (http://www.spotme.ch/spotmeinfo.html). As shown in Figure 5, the Spot-me device is a small mobile device that communicates over radio with other co-located Spot-me devices in the close vicinity of its user. This device is intended to function as an "ice-breaker technology" at conferences or other social gatherings. Using this

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Figure 5. The Spot-me device (left) and a screenshot of the social radar (right)



device, the user can, amongst other things, search for other Spot-me users or use the "social radar" function (Figure 5, right).

The radar function displays the photos and details of all people standing up to 30 meters away (and the range can be altered between one to three meters, seven-20 meters, and 20-30 meters). This functionality provides a completely new way to discretely decide whom to meet during coffee, to personally identify the people sitting nearby at lunch, or to look up forgotten names according to the manufacturer.

Besides these more or less "extreme" interaction devices developed, there has also been a lot of effort made today on developing all various kinds of additional hardware and add-ons to mobile phones to enable us to communicate more easily and in totally new ways using, e.g., our mobile phone digital cameras and video recorders, chat boards, (i.e., tiny SMS/MMS keyboards that can be attached to a mobile phone), digital pens, ringer notification pens, etc.

Interaction in the Wild: Wireless Networking & Mobile Computing

As the previous section has illustrated, mobile computing and wireless networking are two components that enable the Interaction Society. These two technologies make us more or less independent of the current location. With wireless and mobile technologies we can bring the technology along and communicate with anyone, from any desired location, no matter if you are sitting in your office or gone hiking "in the wild."

Although wireless networking and mobile computing are often related, they are not identical as Table 1 illustrates. In this 2× 2 matrix we see a distinction

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Table 1. Mobile vs. wireless diagram (based on Tanenbaum, 2002)

		Mobile	
		No	**Yes**
Wireless	**No**	Desktop computers in offices	A notebook computer used in a hotel room
	Yes	Networks in older, unwired buildings	Portable office; PDA for store inventory

between fixed wireless and mobile wireless. Even notebooks computers, or laptops, are sometimes wired. For example, if a traveler plugs a notebook computer into the telephone jack in a hotel room, he or she has mobility without a wireless network (Tanenbaum, 2002).

On the other hand, some wireless computers are not mobile. An important example is a company that owns an older building lacking network cabling, and which wants to connect its computers without having workmen put in cable ducts to wire the building which also might be a much more expansive solution (Tanenbaum, 2002).

In summary, the wireless technologies make us more independent in that we can now unplug the wires to our machines. Also, the fact that our computers are now quite small, lightweight and movable (i.e., mobile) enable us to bring our computers along wherever we want to go.

A good example of a combination of mobile computing and wireless networking is the mobile phone. The mobile phone is in fact a small mobile computer (i.e., a carry on mobile computational device) with a wireless network connection. Today, the mobile phone has evolved into a quite personal device and, as such, it comes in a lot of different forms with different functionality. However, although the form of the mobile phone and some added functionality might vary a lot from one phone to another (for an example of the difference in

Figure 6. Three Nokia mobile phones (left) and three Ericsson mobile phones (right)



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the shape of modern mobile phones see Figure 6), they are all built to support one fundamental human activity, i.e., human-to-human interaction.

Applications for Interaction or Interaction Software

It is not solely the wireless networks, the Internet infrastructure, or the mobile interaction devices that enable the Interaction Society. Another important enabling component for the Interaction Society is the wide adoption of various new applications for interaction or "interaction software."

Already in the early 1990s people started to, on a widespread basis, use computers to support their everyday communication needs. The most common application for Internet communication around 1994-95 was email, which is still the most widespread channel for Internet communication today, 10 years later[3]. Besides email, it was quite common, and still is, to discuss issues and interact over the Internet via Newsgroups, various kinds of discussion forums, as postings on public virtual bulletin boards, or via different kinds of open chat rooms.

A couple of years later, or more precisely in 1996, the first instant messaging system called ICQ ("I Seek You") was released[4]. With this piece of software installed, people can send short messages to each other in a peer-to-peer fashion.

The ICQ client had, and still has, one big difference from the email systems. The design of ICQ was centered around the *buddy list*, i.e., a dynamic contacts list that continuously and synchronously shows who is currently online, whereas the email system was centered around the list of messages sent and received and ways of storing messages in different folders, etc.

Another central aspect of ICQ was that since it was focused on maintaining social contacts and the focus was set on the persons and the maintaining of the person's social network rather then on the messages *per se* a typical ICQ session typically includes many short messages (almost like post-it notes) sent back and forth between two persons in a chat-like fashion (i.e., almost synchronous interaction) rather then being like email conversations where people sometimes send several pages long messages. In fact, already in 1994 a research project at AT&T Labs Research conducted some experiments with a system called TeleNotes (Whittaker et al., 1997) that was designed to support brief ICQ-like interactions over a network across any geographical distance.

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The original idea behind ICQ was quit simple, but totally in line with the central claim in this book, i.e., that one of the foremost things that people want to do with the technology today is to communicate with one another. As stated on the ICQ website:

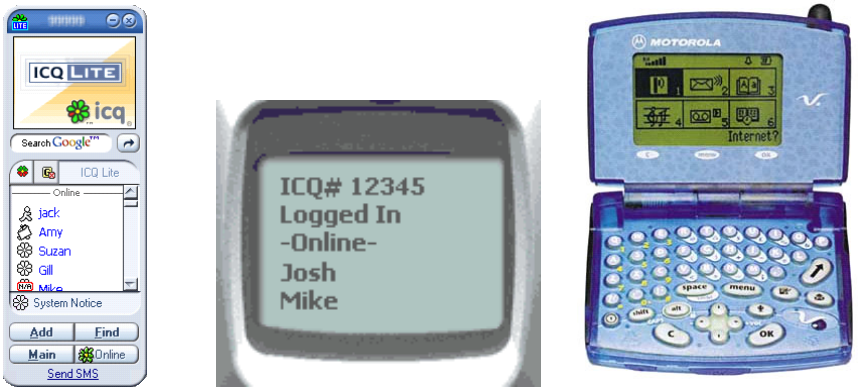
> *"ICQ Inc., the successor of Mirabilis Ltd. was created when America Online acquired all Mirabilis' assets on June 1998. Mirabilis was founded in July 1996 when four young Israeli avid computer users established a new Internet company. Yair Goldfinger (26,Chief Technology Officer), Arik Vardi (27,Chief Executive Officer), Sefi Vigiser (25,President), and Amnon Amir (24, currently studying), created the company in order to introduce a new way of communication over the Internet. They observed the fast deployment of the World Wide Web which was propelled by the mounting popularity of surfing and browsing, and watched the growing number of people interacting with web servers. They realized, however, that something more profound was evolving under the surface. Millions of people have been connected to one huge world wide network — the Internet. They noticed that* ***those people were connected — but not interconnected.*** *They realized that if one missing component would be added, all these people, in addition to* ***interacting with web servers, would be able to interact with each other.*** *The missing link was the technology which would enable the Internet users to locate each other online on the Internet, and to create peer-to-peer communication channels, in a straight forward, easy, and simple manner. They pioneered this technology, that way opening a whole new industry." (http://company.icq.com/info/icqstory.html)*

Nowadays, and in line with the development of not only the Internet infrastructure, but also the mobile phone network, people are also provided with mobile access to ICQ and similar instant messaging systems to meet the communication demands from today's mobile users. Today, ICQ can be run on not only stationary and laptop computers, but also on PDAs and even on mobile phones using WAP (Wireless Application Protocol) or SMS. Figure 7 shows three different ways of accessing ICQ.

Of course, people also still communicate frequently using less sophisticated software than ICQ. Today, several million SMS (Short Message Service) text

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Figure 7. The three pictures illustrate three different ways of accessing ICQ (The first picture (left) shows the PC ICQ Lite client. The screenshot shows the buddylist in the middle of the interface where the user can easily see who is currently online and easily send them short messages by just clicking on a contact in the list. The second picture (middle) shows the mobile phone (WAP) interface to ICQ. Finally, the last picture (right) shows a mobile device (i.e., a Motorola chat phone) where the user can communicate with other ICQ users using simple SMS commands.)



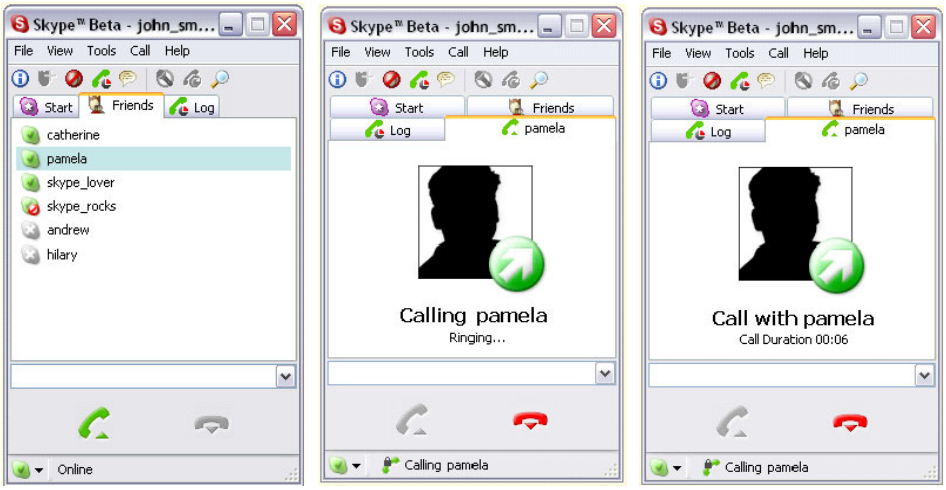
messages are sent everyday worldwide, and people have also during the recent year started to adopt the new format MMS to send short texts, sound clips, and digital photos to each other using their mobile phones with tiny built-in digital cameras.

Another promising area for online communication is IP telephony over the new Internet protocol IPv6. With IP telephony people can have voice communication across IP telephones, ordinary analog telephones, mobile phones and PCs equipped with IP telephony software and a headset.

A special form of IP telephony has just recently been available in the last year, i.e., IP telephony over the Internet using true P2P (peer-to-peer) technology. With this system persons can communicate in full duplex voice mode with each other independent of any central administration of the system. The system for this new kind of IP telephony is a freeware program called Skype and was released in 2003. Although it has only been around for a year it has already more than 7 million users worldwide. Skype is a free and simple software that

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Figure 8. Three screenshots of the Skype IP telephone PC client (The first screenshot (left) shows a buddy list where the green icons indicating whom else that is currently online. If the user wants to make a phone call to someone else that is currently online (e.g., to call Pamela) he or she can simply click on that person in the buddy list to initiate a call (middle). During the call (right) the user can communicate via voice and when the user wants to end the call he/she can simply push the red hang up icon.)



will enable its users to make free phone calls from their PC anywhere in the world (see Figure 8). Skype was created by the people who developed KaZaA (www.kazaa.com) and it uses innovative P2P (peer-to-peer) Internet technology to connect the Skype users worldwide.

Besides this P2P trend there are also a lot of research efforts taken in several new directions to support new forms of human interaction. For instance, there is some interesting research going on to design, e.g., persistent conversation systems (e.g., Erickson et al., 1999; Smith & Fiore, 2001), interaction initiation support (e.g., Wiberg, 2002), stranger interaction support (e.g., Reingold, 2003) and various kinds of awareness systems (e.g., Gutwin & Greenberg, 1996, 1998), and, e.g., ambient displays for informal social events (e.g., Ishii et al., 1998; Mankoff et al., 2003).

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New Interaction Modalities

Another component that has played an important role in enabling the Interaction Society is new ways of interacting *with* these new devices and software available, and new ways of putting the technology to use, i.e., new interaction modalities. The most common of these new interaction modalities is the ability to bring the technology along, i.e., the *mobility* of the technology. This has been possible due to development of better batteries, small graphical displays, tiny microprocessors, and small low-power hard disks. Another important factors in relation to these mobile devices including PDAs and mobile phones is the development of new input devices such as the touch sensitive display and the stylus pen for handwritten input.

Also, the possibility to now bring the technology along without occupying our hands is an emerging area under the name "*wearable computing*" where the basic idea is that instead of carry the devices in our hands it should be worn in, e.g., our clothes, on our back, on our shoulders or even on our head.

In line with this recent "wearable computing" trend there has been some interesting research conducted with a specific focus on what philosophical questions that this new technology realm touches upon (see, e.g., Dourish, 2001; Fällman, 2004; Lund, 2003). Amongst the most popular ideas right now is that this trend towards wearable computing, and also efforts made at realizing so-called "tangible interfaces" (e.g., Ishii & Ullmer, 1997; Brave et al., 1998) is something that very actively and directly involves not only our heads and our fingers but our whole human bodies in our interaction with and via computers. Further on this computing paradigm, where our bodies get more involved and where the augmentation of our experiences has been pinpointed as a central aspect of this phenomenon, some researchers (e.g., Dourish, 2001; Fällman, 2004) have argued that this might be best understood and approached from a phenomenological perspective. All these efforts made on involving our whole bodies (and not just our heads) in our interactions with and via computers have recently been labeled by Dourish (2001) as "embodied interaction."

Another important part of the development of new interaction modalities is all the efforts taken to make our computers visually disappear into the surrounding; disappear and offload our perception and let us focus on the issues and activities we want to focus on instead of forcing us to focus on the technology *per se* whenever we need computer support. Here, important research has recently been conducted on how to design new interaction systems based on embedded technology, the ubiquitous computing ideal and pervasive comput-

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ing paradigm, and current ideas about so-called ambient intelligence where, e.g., peripheral displays have been designed to support informal communication (e.g., Mankoff, 2003). One fundamental idea behind this "hide the computer" ideal has so far been to explore how various kinds of sensors (e.g., touch sensors, light sensors, accelerometers, etc), agents, and other context-awareness technologies (e.g., GPS positioning, WLAN triangulation, etc.) can be used in the design of new interaction technologies.

New Interaction Networks: Local (Spontaneous) Networks

A final component that enables new ways of interacting in our modern society that has just recently started to become widely adopted are various kinds of local open wireless networks of different kinds. These networks have sometimes been labeled "spontaneous" networks since they typically support ad-hoc networking, which is good if somebody wants to quickly establish a network connection for a short while or if the user is at a new location and does not know if there is any network connection available in the ether.

These spontaneous networks that open up new possibilities for people to interact in new ways include, e.g., WLANs (wireless local area networks), ad-hoc networks, Bluetooth networks, and P2P (peer-to-peer) networks.

New Behaviors: An Emerging Society on all Different Levels of Computer Supported Human Activities

The previous sections above have pointed at some important enabling components for the Interaction Society. One might think that "that's just technology." Technology has, however, always had an important influence on the shaping of our society and, on the other hand, our society has had great influence on the shaping of our technology and our technology use[5].

The general hypothesis put forward in this book is that the current transformation from the *Information Society* as described by, e.g., Webster (2002), enabled by the *Network Society* (Castells, 1996), to the *Interaction Society*,

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touches upon, and has some severe impact on several dimension of the modern world, including issues related to individuals, groups, organizations, and the society at large. It also spans across different problem areas of research related to social and cultural issues, technical issues, and theoretical challenges. It also spans across work related interaction, leisure-time interaction, and other forms of social interaction.

To just illustrate the impact of the Interaction Society in how it leads to new behaviours related to the different levels of human activities as mentioned above, we can for example see how new information and communication technologies (ICTs), and interaction applications and interaction devices have implications for *individuals* in their work environments on, e.g., how they should make up their priorities between individual tasks and interaction with others (i.e., a question of interactiveness vs. interpassiveness). In this book, Kakihara et al. explore exactly this specific issue in more detail and in relation to the fluidity of work in organizations.

On a *group level*, or collaborative level of human activities, this transformation into the Interaction Society relates to questions concerning mutual awareness, coordination, and division of labor. While a lot of experimental research has been conducted since the mid-1980s around CSCW (Computer Supported Cooperative Work) to explore new ways of supporting groups and team collaboration with new information technologies these experimental efforts are now a reality. Today, people around the world communicate frequently over email and mobile phones, they participate in electronic meeting rooms and share documents in virtual work environments on the Web. What was once solely prototype systems have now left the research labs and are now available everywhere. One such example is, as mentioned in the previous section, ICQ and similar instant messaging applications that are now a technology that is available and adopted on a large-scale basis.

On an *organizational level* of analysis, an increase in interactions and use of computerized interaction support relates to questions concerning, e.g., how to organize work effectively according to this new society, and how to make use of interaction to build up and share competence (which is closely related to current research in the area of knowledge management). This issue has been in focus for some while now and several studies have been conducted on, e.g., how the use of email affects the efficiency of work in an organization, or how instant mesaging could be used in organizational decision processes on a management level.

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Clearly, this new technology has already enabled, and will even further enable, new ways of doing and organizing work. Some researchers in this area have for example pointed at "networking" as a new way of doing business where interpersonal contacts and interaction technologies play important roles to make the network work (Ljungberg, 1997). This networking idea has also been extrapolated to the society as a whole (see, e.g., Castells, 1996) and there are strong reasons to believe that even more new business models will emerge on top of this "networking" trend[6].

On a *societal level* the transformation towards the Interaction Society is also a shift from computing focused on transactions (e.g., administrative systems, banking systems, datawarehouses), towards computing to support interaction (e.g. mobile phones, ICQ, chat, email, etc.) (Ljungberg & Sørensen, 2000; Dourish, 2001). As mentioned above, it almost makes sense to talk about IT as an *interaction technology*, and not merely an *information technology*. A general vision often put forward here is that interaction in the near future will be seamless, effective, "anytime, anywhere," and instant (stretching from various application domains such as the use of SMS and MMS among teenagers to interaction among business executives and use of interaction technologies to enable financial services). Some researchers (e.g., Dahlbom, 1996, 1997) have even argued that our society of today might be best described as a "talk society" where we, by just picking up our mobile phone, can get in contact with anyone at anytime to do business, to negotiate, to make deals, and act as knowledge-brokers and service buyers and consumers on an open and free market.

On this societal level of human activities we have also already started to see new mass-behaviors emerge during the most recent years. One such phenomenon is "flash mobs" (Reingold, 2003) where several hundred mobile phone users gather together for a fraction of time to do some collective action. It might be to form a line of a couple of hundred persons in front of a hot dog wagon for a couple of minutes and then just walk away from the place a couple of minutes later as if nothing has happened, or gather together to do something else for a couple of minutes and then split up again a couple of minutes later. Reingold (2003) has described this phenomenon as "stranger interaction" where total strangers use their mobile phones to get together and quickly organize themselves to do some collective action without knowing anything about each other.

These sections above have just exemplified very briefly what this new technology can do for us and what it does to us. The rest of this book will report on a lot of additional behaviors emerging in the use of these interaction technolo-

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gies and it will go into detail in describing and analyzing these different behaviors.

New Challenges: Research Areas and Design Issues

There are a lot of challenges to the Interaction Society; many issues that need to be explored. In this section, I will only briefly touch upon a fraction of those challenges to set the scene for the rest of this book. Each chapter in this book will then add to this list of issues that needs to be dealt with in the near future.

The challenges for the Interaction Society can be found on all of the different levels of human activities as mentioned above, and these new technologies that enable the Interaction Society raise a lot of social and technological questions, as well as raise some interesting theoretical challenges for further research.

Concerning the *social dimensions* of the Interaction Society, one obvious issue is that with an increasing growth in adoption of technologies to support interaction, and with an increase in interpersonal communication enabled by these devices and gadgets, *interruptions* will be a focal issue when the geographical place is no longer a useful filter for interaction (because of new possibilities of interruptions caused by mobile phones, pagers, PDAs, etc that break right into conversations "anytime, anywhere"). Further on, an increase in computer-mediated communication might lead to issues concerning, e.g., interaction overload (Ljungberg & Sørensen, 2000), divided attention problems (Wiberg, 2001a, 2001b), and stress, both in the workplace and during leisure hours. Here, empirical case studies are important to describe successful arrangements of interaction technologies, and work routines to enable effective, and maybe healthy, fluid work. Further on, there is a need for analysis, models, new knowledge, and design guidelines for how to effectively cope with the increasing burden of interaction demands, and the increasing demands for instant interaction and demands for being "always online, always available."

Concerning the *technical dimensions* of the Interaction Society, some work is needed on how to create integrated environments across stationary and mobile computers, realize service handover, seamless session management, etc., to enable us to focus more on interacting and communicating with other persons and less on the technical aspects like establishing sessions, configuring devices, finding and initiating network connections, etc.

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There are also some interesting *theoretical challenges* in relation to the rise of the interaction society. Some examples here might be, e.g., how to organize these seamless/fluid work environments? Here, models that can better inform us about the consequences and impact of interaction technologies on work life and human performance are highly relevant. We also need to develop our theories of basic human communication that take into account ongoing interaction across multiple devices, and across different media channels. Finally, and in relation to the section above, we also need new models on how to balance the question that the fluidity in work for one person (empowered by this new technology) is a potential interruption for another.

This book is a first attempt to start addressing these and several additional issues. Thus, the overall objective and mission of this book is to provide its audience with a rich overview of the emerging *Interaction Society* enabled by new information and communication technologies (ICT), including gadgets such as mobile phones, Love getties, PDAs, and pagers, and applications such as email and chat clients, Internet communities, instant messaging systems, video conferencing systems, and different kinds of alert- and notification systems.

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Endnotes

[1] At present (i.e., August 28, 2003), 1.2 billion people worldwide own a mobile phone according to HS Business & Finance (http://www.helsinki-hs.net/news.asp?id=20030828IE7) and to just give an example the total number of telephone users in China has now exceeded 287 million, including 167 million fixed phone users and 120.6 million mobile phone users. China now has the second largest telephone network in the world and it ranks third in the world in terms of information industry. China's number of mobile phone users has just narrowly surpassed the United

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States, whose mobile phone users number 120.1 million. This according to People's daily (http://english.peopledaily.com.cn/200109/04/eng20010904_79296.html).

2 Almost 10% of the world's population now has access to the Internet, according to Nua.com, the compiler of Internet statistics. Figures for Internet use had grown to 580.78M people by the end of May 2002, up from 407.1M in December 2000. The Nua study indicates that for the first time ever, Europe has the highest number of Internet users in the world. There are now 185.83M Europeans online, compared to 182.83M in the U.S. and Canada, and 167.86M in the Asia/Pacific region.

3 According to a research project at Berkeley university, US that conducts studies of the development of email (http://www.sims.berkeley.edu/research/projects/how-much-info/internet/emaildetails.html) the total number of electronic mailboxes in the world had soared 83.5% in the past year (i.e., 2000) to 569,171,660 mailboxes; In the U.S., in the year 2000 the number of mailboxes has jumped 73% to 333.5 million mailboxes since the end of 1998. In the rest of the world, the total number of mailboxes has grown 101% to 235.6 million mailboxes in 2000. In the U.S., the average corporate email user has around 1.5 mailboxes, and the average household using email has about four mailboxes. In the year 2000 there where about 89 million Americans using email at work and roughly 50 million households using email.

4 Source: http://company.icq.com/info/icqstory.html

5 For a more detailed discussion of how technology and society is, and have always been, heavily and complexly intertwined (see, e.g., Castells, 1996).

6 Although the very concept of "networks" or "networking" has recently become very popular, the idea of "networks" or even "knowledge networks" is rather old. For a throughout discussion of, e.g., networks for knowledge creation and sharing from the perspectives of Leibniz and Hegel see Churchman (1972).

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